Introduction to Data Science
Midterm Project

Apply data preparation steps (which can be applied) and calculate descriptive statistics for the any data set. In this project, we are going to use a modified version of Dataset which can be downloaded from the following link.
https://vincentarelbundock.github.io/Rdatasets/articles/data.html

The following steps to follow:

1. Download any data set. Must have numerical and categorical features(more than 10 Features).
2. Delete some features values(must have missing value) for missing value handle. Two ways to handle missing values:
   i. Discard Instances
   ii. Replace by Most Frequent(categorical)/Average Value(numerical)
3. Add Noise value and check noisy value.(categorical and numerical) and handling noise data
4. Check invalid data.
5. Data normalization and transformation.
6. Check Data outlier
7. Descriptive statistics(plot)
8. Data set split(Training 80%, testing 20%)
9. Feature Selection(classification or regression model)

Instructions:

1. Submit the **implemented R program (R file) and updated dataset** in the google form(**Introduction to Data Science**).
2. During VIVA session, you will bring this implemented program and we may ask you to execute the program.
3. **Submit a report** as well. See the instruction section below for the report details. Please bring the printed copy of the submitted report during the VIVA session.
4. At the beginning of the report, write a short note about the dataset. You will get the dataset details from the above link provided for the dataset.
5. For each implemented code segment in the R program, provide the code and its output along with their description in the report.
6. In the description part, only write the content (do not write unnecessary content) that is sufficient to understand the code and its output.
7. **Comments are not allowed in the R program.**